

1 First-Order Secure AES Implementation (zero-randomness variant)

1.1 Metrics in Application-Specific Integrated Circuit (ASIC)

For the ASIC synthesis, we used the Synopsis Design Compiler v2016.03 with the NanGate 45nm Open Cell Library. For performance evaluation, measuring latency and maximum clock frequency (FMAX) is essential. By incorporating the block size (BSIZE), maximum throughput (THR) can be calculated, as indicated by Equation (15) [1].

$$THR = (FMAX \times BSIZE) \div LAT. \quad (1)$$

In our masked S-box implementation, the area is 2686 GE, and the latency (LAT) is 6 clock cycles. In our masked AES implementation, the area is 7921 GE, and the latency is 236 clock cycles. Then, we calculated the ratio of throughput to area. Their detailed performance is shown in Table 1. In contrast to these implementation schemes [2–6], our scheme does not require online fresh randomness and has advantages in terms of latency. Without fresh randomness, both Sugawara’s scheme and Askeland et al.’s schemes employ the changing of the guards [7, 8]. Compared to Sugawara’s scheme, our scheme demonstrates superior area efficiency and lower latency. In contrast to the low-area scheme proposed by Askeland et al., our scheme reduces area consumption by 459 GE and decreases latency by 6 clock cycles, and reduces delay by 0.27ns. Moreover, compared to the free fresh randomness TI of AES proposed by Shahmirzadi and Moradi [9], our scheme achieves a 512 GE reduction in area and a 10 clock cycles reduction in latency. our scheme is inferior in terms of delay and throughput/area. However, we emphasize that their design is based on a non-uniform masked S-box, whereas our masked S-box is uniform allowing for a stronger security argument. Consequently, on the ASIC implementation platform, our work has the lowest area and latency when compared to various state-of-the-art works that do not use online fresh randomness.

1.2 Metrics in Field Programmable Gate Array (FPGA)

FPGAs are composed of numerous programmable slice (SLC) logic resource units. The resource usage for the masked AES hardware design is obtained from the implementation reports generated by Design Suite 14.7 with Xilinx Spartan-6 (XC6SLX75) FPGAs.

In Spartan-6 FPGAs, the architecture of each slice consists of 4 Look-Up Tables (LUTs) and 8 Flip-Flops (FFs). By utilizing the slice and the throughput, we can calculate the design implementation efficiency (Throughput-per-Slice, THR/SLC) as depicted in Equation (16). This equation describes the relationship between the implementation size and performance [1].

$$THR/SLC = THR \div SLC. \quad (2)$$

Table 1. Performance figures of different (byte-serial) implemetations (with key masking, using Synopsis Design Compiler v2016.03, and NanGate 45nm Open cell library)

Design	No.of Shares	Online Fresh Masks [bit]	Area (S-box) [GE]	Latency (S-box) [cycles]	Area (AES) [GE]	Delay (AES) [ns]	Latency (AES) [cycles]	THR/ Area (AES) [Mbps/GE]
[2] ¹	3	48	—	—	11114	—	266	—
[3] ¹	3	44	—	—	9102	—	246	—
[4] ¹	3	44	—	—	11221	—	246	—
[4] ¹	3	32	2840	4	8119	—	246	—
[5] ¹	2	18	—	—	7100	—	246	—
[7] ²	3	0	3500	4	17100	—	266	—
[9] ^{2,3}	2	1	—	6	7619	0.25	246	0.27
[9] ^{2,3}	2	0	—	6	8183	0.29	246	0.21
[8] ²	2	0	2710	6	8130	1.21	242	0.05
[6] ²	2	4	—	6	7843	0.94	236	0.07
this work ²	2	0	2686	6	7671	1.10	236	0.06

¹ Using UMC 180nm Standard Cell Library.² Using NanGate 45nm Standard Cell Library.³ with a non-uniform masked S-box.

In our masked AES implementation, the area size is 212 slices, the latency is 236 clock cycles, and the maximum clock frequency is 125 MHz. Additionally, we achieved a maximum throughput of 67.8 Mbps, resulting in a Throughput-per-Slice of 0.319 Mbps/slice. The implementation details are provided in Table 2. Consequently, on the FPGA implementation platform, when compared to various state-of-the-art works that do not use online fresh randomness, our work also has the best implementation efficiency (THR/SLC).

Table 2. Comparison of area-optimized (byte-serial) implementations including PRNG (with key masking, mapped on Xilinx Spartan-6)

Design	SLC	FF	LUT	BRAM	LAT [cy- cles]	Online Fresh Masks[bit]	FMAX [MHz]	THR [Mbps]	THR/ SLC [Mbps/slice]
[5]	433	726	894	0	216	28	110	65.1	0.150
[5]	351	754	792	0	246	18	134	69.7	0.198
[10]	418	690	905	0	276	54	133	61.6	0.147
[11]	254	124	347	0	6852	18	103	1.9	0.007
[12]	140	92	248	0	6852	6	120	2.2	0.015
[9]	631	866	1021	0	246	0	136	70.7	0.112
[13]	203	529	476	108kb	216	8	138	81.7	0.402
[6]	213	587	700	0	236	4	108	58.6	0.275
this work	212	562	561	0	236	0	125	67.8	0.319

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